

The nonsolvable groups of order 15 360

Cross-checks by the Zphi and supplement method

Claude Archer – code for the Zphi GAP package

Extensions of Alt(5) by groups of order 256

#grps	$\mathcal{O}_K$	total
56 092	1 027 380	1 083 472

Extensions of SL(2,5) by groups of order 128

$\mathcal{O}_Z$	$\mathcal{O}_{Z,K}$	total
337 956	6 856 498	7 194 454

Extensions of perfect groups of order 960

P	total
$P_1 = \text{PerfectGroup}(960, 1)$	141
$P_2 = \text{PerfectGroup}(960, 2)$	42
Total	183

Ext. of centerless groups (GAP-grpconst : UpwardsExtensionsNoCentre, B. Eick).

Extensions of perfect groups of order 1920

P	total
$P_1 = \text{PerfectGroup}(1920, 1)$	261
$P_2 = \text{PerfectGroup}(1920, 2)$	261
$P_3 = \text{PerfectGroup}(1920, 3)$	196
$P_4 = \text{PerfectGroup}(1920, 4)$	34
$P_5 = \text{PerfectGroup}(1920, 5)$	63
$P_6 = \text{PerfectGroup}(1920, 6)$	63
$P_7 = \text{PerfectGroup}(1920, 7)$	63
Total	941

Extensions of perfect groups of order 3840

P	total
$P_1 = \text{PerfectGroup}(3840, 1)$	40
$P_2 = \text{PerfectGroup}(3840, 2)$	40
$P_3 = \text{PerfectGroup}(3840, 3)$	60
$P_4 = \text{PerfectGroup}(3840, 4)$	131
$P_5 = \text{PerfectGroup}(3840, 5)$	240
$P_6 = \text{PerfectGroup}(3840, 6)$	22
$P_7 = \text{PerfectGroup}(3840, 7)$	68
Total	601

PerfectGroup(3840,6): see Theorem 3.2.1 in C. Archer, pg 40–102.

Order 15 360

Extensions of perfect groups of order 7680

P	total
$P_1 = \text{PerfectGroup}(7680, 1)$	11
$P_2 = \text{PerfectGroup}(7680, 2)$	10
$P_3 = \text{PerfectGroup}(7680, 3)$	10
$P_4 = \text{PerfectGroup}(7680, 4)$	16
$P_5 = \text{PerfectGroup}(7680, 5)$	12
Total	59
Total PerfectGroups(15360)	7

Total non solvable groups of order 15360

family	total
Extensions of Alt(5) by groups of order 256	1 083 472
Extensions of SL(2,5) by groups of order 128	7 194 454
Extensions of perfect groups of order 960	183
Extensions of perfect groups of order 1920	941
Extensions of perfect groups of order 3840	601
Extensions of perfect groups of order 7680	59
Total PerfectGroups(15360)	7
Global total	8 279 717